Shiv Nadar University

Undergraduate Course Description Form

Semester: IV (Spring 2019)

PART A: COURSE IDENTIFIERS

1. School	**School of Engineering**
2. Department	**Computer Science and Engineering**
3. Course Code	**CSD202**
4. Course Title	**Database Management Systems**
5. Credits (L:T:P:C)	**3:0:1:4**
6. Contact Hours (L:T:P)*	**3:0:2**
7. Prerequisites	**CSD201, CSD205/MAT140**
8. Major Core for**	**CSE**
9. Major Elective for	**None**

***Per student per week. For example, a 1.5 credit CCC that runs for a half-semester with only lectures would have contact hours 3:0:0. A course with 3 lecture hours and 1 tutorial hour and a single 3-hour lab per week would have contact hours 3:1:3.
**Give Major and Year.

PART B: OBJECTIVES AND PRACTICE

10. Course Summary

This course is designed to equip students with knowledge about the fundamentals of Database Management Systems. The course also has a significant lab component. Through this lab component, students will gain exposure to SQL as well as procedural SQL. Topics include introduction to DBMS (Database Management Systems), ER model, relational data model, relational algebra, normalization, indexing, query processing & optimization, transaction processing, concurrency control & recovery, and an introduction to some advanced topics such as data mining, data warehousing and Big Data.

11. Course Aims

- To equip students with knowledge about the fundamentals of Database Management Systems
- To enable students to program in SQL and procedural SQL.

12. Learning Outcomes

On successful completion of the course, students will be able to achieve the following:

1. Understand database architecture and database system environment
2. Understand data models and schemas

3. Write database requirements analysis for building a DBMS for a given organization or subset of an organization
4. Design ER model to satisfy database requirements
5. Design a relational database using SQL, while satisfying relational model constraints
6. Understand the importance of indexing
7. Design indexes based on application requirements
8. Understand query processing & optimization, while taking trade-offs into consideration
9. Understand transaction processing, concurrency control & recovery
10. Write programs in SQL

13. Curriculum Content

No. of Weeks	Topics	Lab
1	Introduction to Database Management Systems (DBMS), database architecture, data models, schemas, database system environment, classification of DBMS	Database requirements analysis: determine the database requirements of a given organization (e.g., a company) or a part of the organization (e.g., library)
1.5	Entity Relationship (ER) model	Create ER model for the previous assignment
5	Relational data model concepts, relational model constraints, relational database schemas, triggers, views, relational algebra, SQL, Normalization and functional dependencies for Relational DBMS	SQL assignment: Create a relational database for the above. The database should satisfy the relational model constraints. Normalize upto BCNF. Update the database; issue queries (including range queries) on the database. SQL Lab test: Testing whether the students have reasonably good knowledge of SQL
2	indexing, B-trees and its variants	Implement a B-tree or one of its variants queries including range queries, insert and update functionalities should be supported; should also incorporate support for various kinds of databases.
1	Query processing and query optimization	
4	Transaction processing, concurrency control and recovery	
1	Overview of data mining, data warehousing, Big Data	

14. Teaching and Learning Strategy

All materials (including lecture slides, assignments, labs, etc.) will be uploaded on BlackBoard. Please refer to the modules in Blackboard for details.

PART C: ASSESSMENT.

15. Evaluation Instruments

Evaluation Instrument	Weightage	Time	Comment
Mid-term Exam	20%	1.5 hours	In Class
Assignments/Quizzes	15%	By due date/time	
Laboratory and Project	35%	By due date/time	In Class + Homework
End-term Exam	30%	2 hours	In Class

16. Mapping of Learning Outcomes to Assessment Strategy

For detailed mapping, please refer to each course module in Blackboard for details.

17. Bibliography

The recommended textbook for the course is:

Fundamentals of Database Systems (7th Edition), Ramez Elmasri and Shamkant B. Navathe, Publisher: Pearson (June 30, 2017); ISBN-10: 933258270X; ISBN-13: 978-9332582705
https://www.amazon.in/Fundamentals-Database-System-Elmasri-Ramez/dp/933258270X?tag=googinhydr18418-21&tag=googinkenshoo-21&ascsubtag=6e74beab-229b-421a-a2bd-f296100c019b

Reference book:

Database System Concepts (7th Edition), Avi Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill, ISBN 0-07-352332-1